

# Social Science Inquiry II

Week 8: Statistical power, effect sizes, sample size

Molly Offer-Westort

Department of Political Science,  
University of Chicago

Winter 2026

# Today's questions

- What is power?
- What drives it?
- How do we justify sample size?

# Learning goals

- Define power and connect it to false negatives.
- Identify the main ingredients that determine power.
- Explain what makes a power analysis believable (or not).
- Write a defensible sample-size justification.

# Power: signal vs. noise

Coppock (2022)

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## Coppock (2022)

- **Power** is the ability to distinguish signal from noise. Formally: the probability of rejecting the null hypothesis when the alternative is true (i.e., detecting a real effect).
- Signal: the treatment effect (or relationship) we care about.
- Noise: how much outcomes vary for other reasons.
- High noise + small effects  $\Rightarrow$  hard to detect.

## Errors you care about

|               | Reject $H_0$                   | Fail to reject $H_0$      |
|---------------|--------------------------------|---------------------------|
| $H_0$ is true | Type I error ( $\alpha$ )      | Correct                   |
| $H_A$ is true | Correct (power = $1 - \beta$ ) | Type II error ( $\beta$ ) |

- Low power means a high chance of missing real effects.

# Normal critical values and z-quantiles

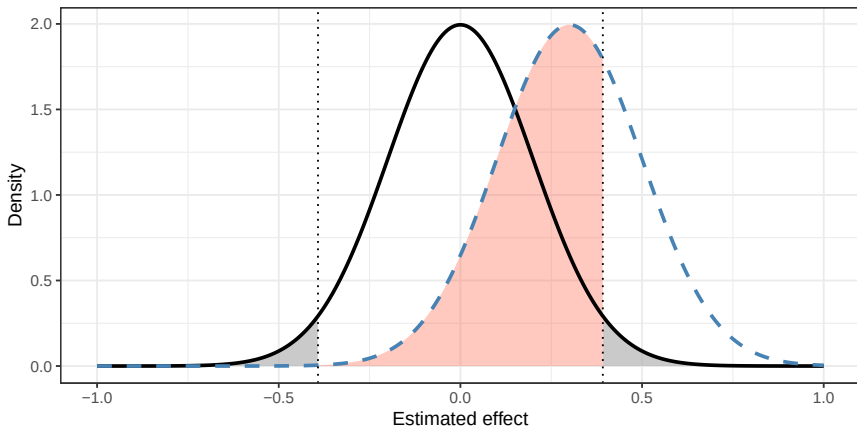
- Under a normal approximation, test cutoffs are measured in “standard error units.”
- Notation:  $z_p$  is the  $p$ th quantile of  $N(0, 1)$  (so  $\Pr(Z \leq z_p) = p$  for  $Z \sim N(0, 1)$ ).
- Common values:  $z_{0.975} \approx 1.96$  (two-sided  $\alpha = 0.05$ ) and  $z_{0.80} \approx 0.84$  (power = 0.80).



# Visualizing $\alpha$ , $\beta$ , and power

Sampling distributions under  $H_0$  and  $H_A$

Two-group design; n per group = 50, sd = 1, delta = 0.3, alpha = 0.05, power ... 0.32



- Gray tails under  $H_0$ : Type I error ( $\alpha$ ).
- Red region under  $H_A$ : Type II error ( $\beta$ ). Power =  $1 - \beta$ .

# Why we need power analysis

- If an effect exists, will our study design be able to detect it?
- Underpowered studies are costly and uninformative.
- Power analysis forces us to make our assumptions explicit:
  - How big is the effect we care about?
  - How noisy is the outcome?
  - What sample size and design can we actually run?

## Quick check

If power = 0.30, what does that mean?

# The three ingredients of power

- **Effect size:** how big is the signal?
- **Noise:** how variable is the outcome?
- **Sample size:** how much information do we collect?
- Design choices also matter (assignment ratio, clustering, blocking) and estimation choices (estimator, covariate adjustment).

# Effect size isn't "what you hope"

- The effect size used in power analysis should be tied to a claim.
- Often, the key question is: **what is the smallest effect size of interest?**
- If we don't define what effects matter, "powered" doesn't mean much.

# Noise comes from measurement and design

- Outcome variability (heterogeneity in the population).
- Measurement error and survey noise.
- Timing (when outcomes are measured).
- Noncompliance and attrition.
- Dependence (clusters, repeated observations).

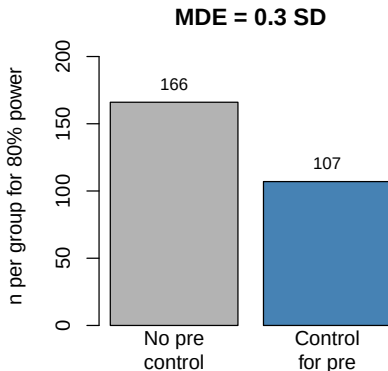
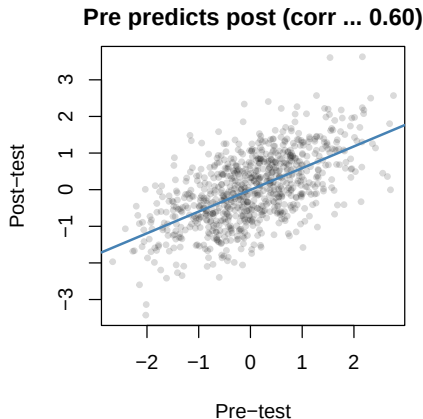
# N is not the only lever

- Increasing  $N$  increases power, but:
- You can sometimes *increase precision* without increasing  $N$ :
  - better measurement (reduce noise)
  - better design (blocking/stratification)
  - better analysis plan (pre-specified covariates)

# Example: pre-test covariates can boost power

Offer-Westort et al. (2026)

- If a pre-test strongly predicts the post-test, controlling for it reduces residual noise.
- Less noise  $\Rightarrow$  smaller  $n$  for the same effect size of interest and target power.





# Applications and tradeoffs

1. For each case, what is the likely bottleneck (effect size, noise, or  $N$ )?
2. What is **one** design change to increase power?
  - Case A: outcome is “monthly income”; small expected effect;  $N$  is limited.
  - Case B: outcome is a low-variance administrative count; moderate effect;  $N$  is moderate.

# Assumptions

- A power calculation maps assumptions  $\Rightarrow$  probability of detection.
- It's a *design tool* to tell us what to do *under certain assumptions*—
- If the assumptions are wrong, the power calculation can be very misleading.

# Minimal detectable effect (MDE)

- Fix  $\alpha$  and a target power (e.g.,  $\alpha = 0.05$ , power = 0.80).
- Given  $N$  and noise, compute the smallest effect you can reliably detect.
- Alternative reframing: **what can this study rule out?**

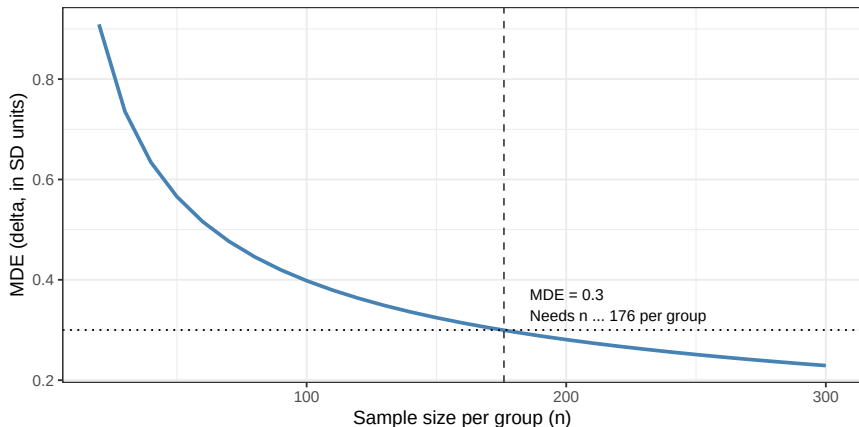
```
alpha <- 0.05
power_target <- 0.80
n_per_group <- 50
sd_y <- 1
mde <- power.t.test(n = n_per_group, power = power_target, sd = sd_y,
                    sig.level = alpha, type = "two.sample",
                    alternative = "two.sided")$delta
cat(sprintf(
  "alpha=%.2f, power=%.2f, n per group=%d (N=%d), sd=%.1f => MDE = %.2f SD\n",
  alpha, power_target, n_per_group, 2 * n_per_group, sd_y, mde
))

## alpha=0.05, power=0.80, n per group=50 (N=100), sd=1.0 => MDE = 0.57 SD
```

# MDE decreases as $n$ increases

With fixed noise + alpha, bigger samples detect smaller effects

alpha = 0.05; target power = 0.80; sd = 1



# When to believe your power analysis

- **Effect size inputs** are justified.
- **Noise inputs** are justified (pilot data, benchmarks, or conservative assumptions).
- The calculation matches the **design and analysis** you will actually run.

# Common failure modes

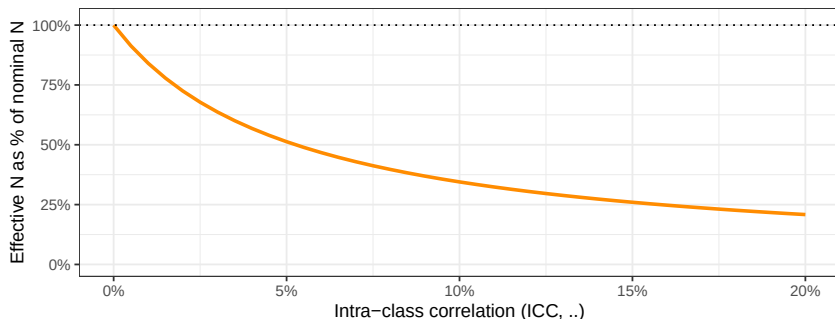
- Using an optimistic effect size from prior literature (especially small studies).
- Ignoring multiple testing or multiple outcomes.
- Ignoring clustering / dependence.

# Clustering: the “effective $N$ ” can be much smaller

- If outcomes are correlated within clusters (schools, villages, classrooms), you have less information than  $N$  suggests.
- Approximation: design effect  $DEFF = 1 + (m - 1)\rho$  where  $m$  is cluster size and  $\rho$  is the ICC.
- Effective sample size (roughly)  $N_{\text{eff}} \approx N/DEFF$ .

Even small ICCs can substantially reduce effective sample size

Example cluster size  $m = 20$ ;  $N_{\text{eff}} / N = 1 / (1 + (m-1)\rho)$



## (Optional) Multiple tests and power

- If you adjust  $\alpha$  for multiple tests, power goes down (unless  $N$  goes up).
- Simple example: Bonferroni correction lowers the per-test  $\alpha$ .
- This is one reason preregistration and a clear analysis plan matter.



# Why simulate power?

- Closed-form formulas rely on assumptions that can be wrong.
- Simulation lets you match:
  - your design (assignment ratios, clustering, attrition)
  - your estimator and analysis plan

# Simulation recipe

1. Assume an effect size, noise, and design.
2. Generate data from those assumptions.
3. Run the analysis you plan to run.
4. Repeat many times; power  $\approx$  fraction of rejections.

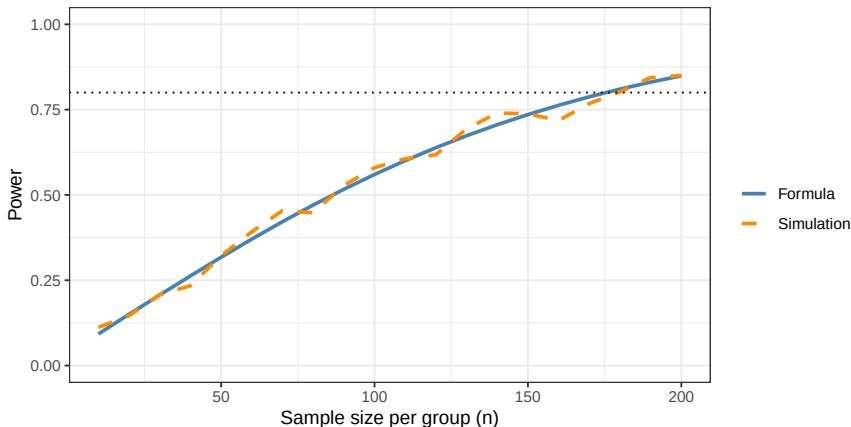
# What simulation buys you

- Handles covariates and complicated estimators.
- Handles noncompliance and attrition (if modeled).
- Handles nonlinear outcomes (binary, counts, etc.).
- Helps you sanity-check your design choices.

# Power curve in a two-group design

Power increases with  $n$  (holding effect size and noise fixed)

Two-sample t-test;  $\alpha = 0.05$ ;  $\delta = 0.3$ ;  $sd = 1$ ;  $sims = 500$



- This is power *under the assumptions used to generate the data.*

# Sample size justification (Lakens)

## Lakens (2022)

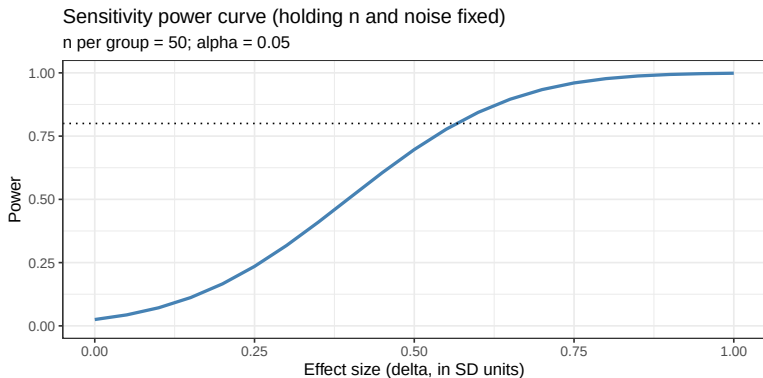
- “Power analysis” is *one* way to justify sample size.
- Other legitimate justifications include:
  - measuring (almost) the full population
  - resource constraints (with transparency)
  - desired accuracy / precision (CI width)
  - heuristics (weaker)
  - explicitly acknowledging no clear justification

# What counts as an “interesting” effect?

- Smallest effect size of interest / minimal detectable effect (MDE)
- Expected effect size (with careful justification)
- Confidence interval width (accuracy)
- Sensitivity power analysis over a range of effects
- Distribution of effects in the research area

# Sensitivity power vs. “powering for the expected”

- “Powered for expected effect” can be fragile if the expected effect is optimistic.
- Sensitivity power asks: across a range of effects, what can we detect?



## Guided example: design + assumptions

- Research question: does a message/frame change support for a policy? Offer-Westort et al. (2026)
- Outcome ( $Y$ ): standardized support (mean 0, SD 1).
- Estimand: difference in means,  $E[Y \mid T = 1] - E[Y \mid T = 0]$ .
- MDE: 0.30 SD (small but meaningful).
- Target:  $\alpha = 0.05$ , power = 0.80, 50/50 assignment.
- Class questions: is this feasible? If not, what lever would you pull first?



## Guided example: compute + justify

```
## n per group (80% power, MDE=0.30): 176 (N=352)
## If n=100, power for MDE: 0.56
## If n=100, MDE (80% power): 0.40
```

- If the feasible  $n$  is too small for your MDE, either change the design (reduce noise) or change the inferential goal (report range of MDEs/sensitivity).

## What you will preregister (Week 9)

- Outcome(s) and estimand(s)
- MDE
- Analysis plan (including multiple testing decisions)
- Planned sample size and the justification approach

In one sentence: what would you change to increase power in your own design?

# References I

Coppock, A. (2022). 10 things to know about statistical power. EGAP.

Lakens, D. (2022). Sample size justification. *Collabra: Psychology*, 8(1):33267.

Offer-Westort, M., Gruen, W., Herron, C., Hyatt, K., Buford, M., Davis, K., Fonseca, D., Gurevich, M., Huang, T., Jerez, R., et al. (2026). Which frame fits? policy learning with framing for climate change policy attitudes. *Research & Politics*, 13(1):20531680251414927.